Title

One line title for this project

Graph Learning for Interpretable Urban Accessibility and Opportunity

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Description

More detailed description of this project (max 500 words).

Cities are complex, interconnected systems where access to opportunities is shaped by spatial structure, infrastructure, and social dynamics. Graph learning offers a powerful framework to model these relationships, enabling interpretable insights into urban accessibility and the distribution of opportunity. The presence of a multimodal data that integrate GTFS transit schedules, sociodemographics (ACS), employment locations (LEDH), and land-use maps (OSM) allow us to to compute accessibility indices (e.g. number of jobs reachable within 30–60 min by transit or car).

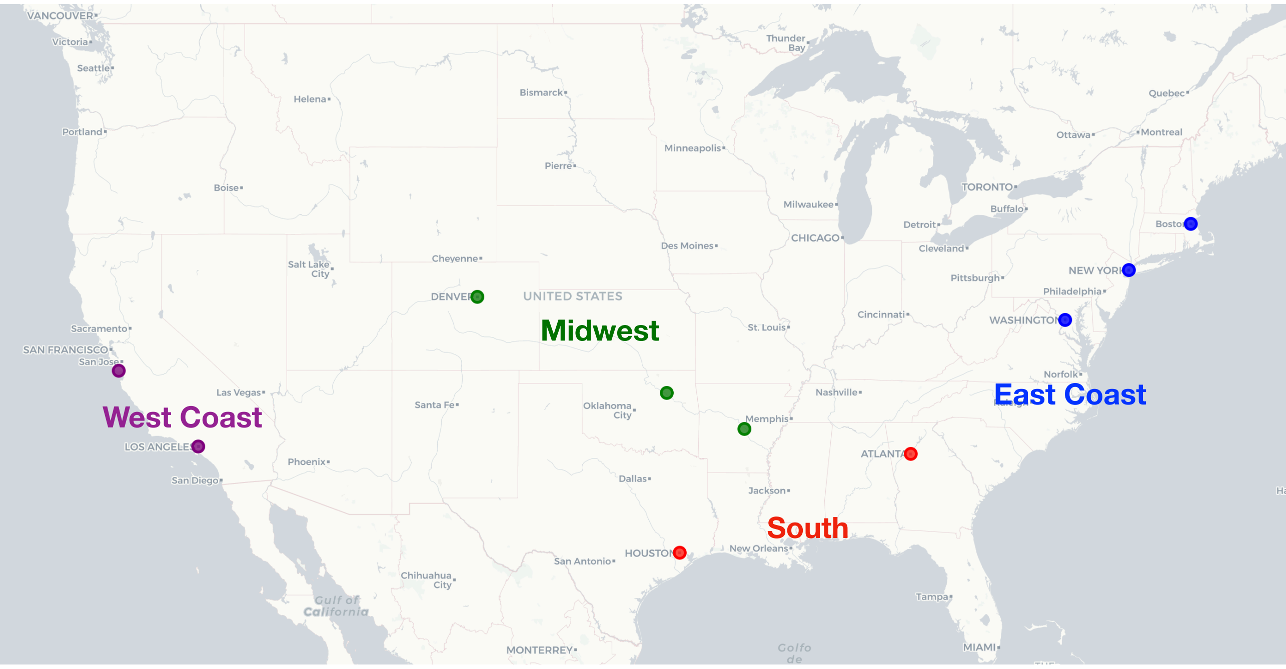
The goal is to train, among others, tree-based graph learning models (Random Forest, XGBoost) to predict these accessibility scores from spatial features (e.g. population density, land-use mix, transit stop density, travel speeds) and then explain the results with SHAP values and partial-dependence plots. Clustering the SHAP "signatures" will reveal zones with similar access patterns. The explainable outputs support equity-focused planning by showing which factors (e.g. lack of transit stops or mixed-use) most depress access in underserved areas.

Within this framework, the project will address two central research questions:

1. How accurately can graph learning models predict multimodal accessibility indices from spatial and socio-economic features across urban environments?
2. Which spatial factors most strongly influence urban accessibility, and how can interpretable methods (e.g., SHAP values and clustering of feature attributions) reveal distinct patterns of opportunity?

Goal

Predicting accessibility (represented by the presence of public transportation) and opportunity (denoted by the availability of workplace) through the lenses of locational context and built environment (captured by distribution of amenities in land-use)



Steps

1. Data Collection:
   - GTFS (https://www.transit.dot.gov/ntd/data-product/2023-annual-database-general-transit-feed-specification-gtfs-weblinks):
     1. MTA New York City Transit (New York City, NY-East Coast)
     2. Massachusetts Bay Transportation Authority (Boston, MA-East Coast)
     3. Washington Metropolitan Area Transit Authority (Washington, DC-East Coast)
     4. Metropolitan Atlanta Rapid Transit Authority (Atlanta, GA-South)
     5. Metropolitan Transit Authority of Harris County (Houston, TX-South)
     6. Denver Regional Transportation District (Denver, CO-Midwest)
     7. Metropolitan Tulsa Transit Authority (Tulsa, OK-Midwest)
     8. Rock Region Metropolitan Transit Authority (Little Rock, AR-Midwest)
     9. Santa Cruz Metropolitan Transit District (Santa Cruz, CA-West Coast)
     10. Los Angeles County Metropolitan Transportation Authority (Los Angeles, CA-West Coast)
   - ACS (https://www.census.gov/programs-surveys/acs/data.html)
   - LEDH (https://lehd.ces.census.gov/data/) —> Workplace Area Characteristics (python: https://lehd.ces.census.gov/data/lehd-code-samples/sections/lodes/basic_examples.html#spatial-and-geographic-analysis)
   - OSM (https://download.geofabrik.de/)
   - TIGER/Line Shapefile (https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html)
2. Data Prepocessing:
   - Public Transportation (GTFS): For each census blocks 15-digit geocode (e.g.: 360010001001003) in each city, identify:
     1. number of census blocks connected by public transportation service from its area
     2. number of bus stops/tram stops/bus stops/stations
     3. total frequency of daily public transportation service
     4. etc
   - Sociodemographics (ACS): For each census blocks 15-digit geocode (e.g.: 360010001001003) in each city, identify:
     1. total population, population by gender, population by ethnicity
     2. average income level
     3. etc
   - Built Environment (OSM): For each census blocks 15-digit geocode (e.g.: 360010001001003) in each city, identify:
     1. number of amenities by category (e.g.: number of restaurants)
     2. total street length
     3. etc
   - Employment Opportunity (LEDH): For each census blocks 15-digit geocode (e.g.: 360010001001003) in each city, identify:
     1. number all the jobs within 1, 2, 3, 4, and 5 km from its centroid
     2. average income level (e.g: median household income)
     3. etc
3. Data Analysis
   - Initialize graph representation where blocks as the node/analytical units, while features related to public transportation, sociodemographics, and built environment serve as node attribute/locational accessibility indicators (see: Xie, et. Al, 2026, p.9)
   - Measure locational accessibility by inverse-weighting correlation coefficients among multimodal data (public transportation, sociodemographics, and built environment) (see: Xie, et. Al, 2026, p.11)
   - Predict locational accessibility using a tree-based ensemble model (e.g.: XGBoost), a Graph Neural Network model (e.g: GraphSAGE), and a Graph Convolutional Network.
   - Run feature importance on each model to capture its local interpretability: XGBoost (use SHAP), GraphSAGE (use GNNExplainer), and GCN (use GNNExplainer).
4. Visualization
   - Design a prototype of dashboard: https://whatif.sonycsl.it/index.html

References

Machine learning for spatial analyses in urban areas: a scoping review (Casali, Aydin, and Comes, 2022)
https://www.sciencedirect.com/science/article/pii/S2210670722003687?utm_source=chatgpt.com
A hybrid spatial learning method integrating graph neural networks and explainable AI to understand urban vitality (Xie, et. Al, 2026)
https://www.sciencedirect.com/science/article/abs/pii/S2210670725007929
Modeling walking accessibility to urban parks using Google Maps crowdsourcing database in the high-density urban environments of Hong Kong (Gong, 2023)
https://www.nature.com/articles/s41598-023-48340-w?utm_source=chatgpt.com